Answer key. Answers are in magenta. Stan results vary a little from computer to computer.

NRES 746, Week 9, Lecture 2: Getting Stan Running (take-home)

Due Monday. We'll start Monday's class by sorting out any installation problems, so bring your laptop. Start the installation early: it downloads and builds a fair amount of software and can take 20-30 minutes, most of it waiting.

Using AI: each coding question is labeled with a recommendation. After each one, circle how you actually used AI. Installation errors can be a good use for an AI assistant (or a web search): paste in the error message and ask what it means.

Part 1: Install Stan

We'll use **cmdstanr**, the R interface to Stan. The full instructions are at https://mc-stan.org/cmdstanr/articles/cmdstanr.html; the short version:

1. **Get a C++ toolchain.** Windows: install Rtools for your version of R (https://cran.r-project.org/bin/windows/Rtools/). Mac: in the Terminal, run `xcode-select --install`.
2. **Install the package and Stan itself**, in R:

```
install.packages("cmdstanr",
                 repos = c("https://stan-dev.r-universe.dev", getOption("repos")))
library(cmdstanr)
check_cmdstan_toolchain(fix = TRUE)    # checks (and tries to fix) the C++ setup
install_cmdstan(cores = 2)             # downloads and builds Stan; be patient
cmdstan_version()
```

1a. What does `cmdstan_version()` report? If the installation failed, write the error message you got stuck on.

CmdStan version: 2.39.0 or newer (any recent version is fine)

(Only if it failed.) Common culprits: Rtools missing or the wrong version (Windows), command line tools not installed (Mac), or a path with spaces or non-English characters.

Part 2: Fit Wednesday's myxo model

Here is the model from Wednesday's lecture (myxomatosis virus titers, Gamma distribution, flat priors on the positive values). Copy it into an R script and run it. Stan compiles the model to C++ the first time, which takes up to a minute.

```
library(cmdstanr)
myxo <- read.csv(
  "https://kevintshoemaker.github.io/NRES-746/data/MyxoTiter_sum.csv")
titer <- myxo$titer[myxo$grade == 1]

myxo_code <- "
data {
  int<lower=0> N;
  vector<lower=0>[N] titer;
}
parameters {
  real<lower=0> shape;
  real<lower=0> scale;
```

```
}
model {
  titer ~ gamma(shape, 1 / scale);  // Stan's gamma uses shape and rate
}
"
myxo_model <- cmdstan_model(write_stan_file(myxo_code))
fit <- myxo_model$sample(data = list(N = length(titer), titer = titer),
                         chains = 4, parallel_chains = 4, seed = 746)
fit$summary()
fit$diagnostic_summary()
```

2a. *(AI optional/encouraged.)* Record these from the output (yours may differ a little from a classmate's):

$\hat{R}$ for shape: 1.00 effective sample size (`ess_bulk`) for shape: about 452

Number of divergent transitions (all chains): 0

95% credible interval for shape (hint: `quantile()` on `fit$draws("shape", format = "draws_matrix")`): about 25 to 76

AI use (circle one): no AI / AI helped (e.g., debugging) / AI wrote the code

2b. *(Recommended: no AI.)* Now make the sampler work much less hard: rerun the `sample()` call with `iter_warmup = 20` and `iter_sampling = 50` added. How do $\hat{R}$ and the effective sample size change, and why would you not trust these results?

$\hat{R}$ for shape: 1.99 (example run; well above 1.01) `ess_bulk` for shape: about 7 (example run)

R-hat rises well above 1.01 and the effective sample size collapses: 20 warmup draws aren't enough for NUTS to tune its step size, and the four chains haven't converged to the same distribution. Summaries from these draws would be unreliable.

AI use (circle one): no AI / AI helped (e.g., debugging) / AI wrote the code

Part 3: Reflection

Where did you use AI, if anywhere, and what did you have to check or fix? If you didn't use it, what was hardest to get working?

Open-ended.